Bayes by Backprop's OOD Advantage Depends on Where Training Stops

A softness-controlled study of mean-field VI along the optimization path

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Abstract

Bayesian neural networks are often said to detect out-of-distribution (OOD) inputs better than deterministic networks. But their predictions are softer everywhere, and a deterministic network can be softened for free by a temperature. I compare each method with a deterministic network softened to the same in-distribution entropy, in a series of pre-registered experiments on MLPs trained on MNIST. At typical training budgets, mean-field Bayes by Backprop (BBB) is no better than this softness-matched network. Its advantage (+0.06 to +0.08 AUROC, more than a deep ensemble) appears only after enough optimizer steps, and the number of steps, not the amount of data, decides it. The posterior standard deviations grow along an almost identical path for training sets of 500 to 10,000 images, suggesting that the optimizer, not the data, sets the pace. Which uncertainty score is used then matters. With predictive entropy the advantage can disappear later in training, because the model also becomes uncertain on in-distribution inputs as it underfits. With mutual information (MI) it persists, but MI also "detects" OOD inputs in runs that never learned the task. Neither held-out NLL nor the ELBO selects a good stopping point, and a simple rule based on synthetic outliers does no better than stopping at a fixed step. On CIFAR-10 with small, underfit models (37–68% accuracy), no comparable advantage appears.

1 Introduction

A standard classifier gives a confident prediction for any input, including inputs unlike its training data. Bayesian neural networks (BNNs) average predictions over a distribution of weights [1], which should raise uncertainty on unfamiliar inputs. The usual evidence is that a BNN's predictive entropy is higher on OOD data than a deterministic network's.

That evidence has a confound. A BNN's averaged prediction is softer than a deterministic network's on every input, and temperature scaling [2] softens a deterministic network at no cost. Temperature is a known ingredient of OOD detection [3] and appears as a baseline in uncertainty benchmarks [4]. Whether BNNs are intrinsically good OOD detectors has been questioned [5], and a Bayesian last layer is known to reduce overconfidence far from the data [6]. The comparison I use throughout is simple: each method against a deterministic network softened, on validation data only, to that method's own in-distribution entropy. The question is whether the method ranks OOD inputs as more uncertain than in-distribution inputs better than a network of the same softness.

Findings.

1. At standard training budgets, mean-field BBB is no better than the softness-matched network on MNIST → Fashion-MNIST and KMNIST. Deep ensembles are slightly better; MC dropout ties; last-layer Laplace is worse (§3.1).

2. BBB's advantage is set by the number of optimizer steps, not by the training-set size (§3.2). The posterior standard deviations follow nearly the same trajectory at every n (§3.4).

3. Along training, the entropy-based advantage can rise and fall while the MI-based advantage persists; MI is also high for models that never learned (§3.3).

4. The advantage needs posterior noise in both hidden layers together, and it is not a pixel shortcut (§3.5).

5. Neither held-out NLL nor the ELBO picks a good stopping point. A synthetic-outlier rule does no better than a fixed step count (§3.6).

6. On CIFAR-10 with small, underfit models, no comparable advantage appears (§4).

Every experiment had its hypothesis, predictions and falsification criteria committed to git before it ran. Appendix B lists all predictions, including the many that failed.

2 Setup

Data. MNIST training subsets of $n \in \{500, 2{,}000, 10{,}000\}$ class-stratified images (one per seed); a fixed validation pool of 5,000 further training images, disjoint from every subset; the MNIST test set; OOD: Fashion-MNIST [7] and KMNIST [8] test sets, normalised with MNIST statistics. CIFAR-10 experiments are described in §4.

Models. All MNIST models use a 784–256–256–10 ReLU MLP. *DNN*: Adam (10^{-3}, cosine), 40 epochs, batch 64. *MAP*: the same network trained to the MAP estimate under BBB's $\mathcal{N}(0, 1)$ prior, with BBB's budget (150 epochs, gradient clipping). *MC dropout* [9]: $p = 0.1$, 50 samples. *Deep ensemble* [10]: 5 DNNs. *Last-layer Laplace* [11]: full generalised Gauss–Newton covariance over the last layer of a DNN, prior precision tuned on validation NLL. *BBB* [1]: mean-field Gaussian posterior, $\mathcal{N}(0, 1)$ prior, per-example negative ELBO

$$\mathcal{L} = \frac{1}{B} \sum_{i \in \text{batch}} -\log p(y_i \mid x_i, w) + \frac{1}{N} \mathrm{KL}\big[q(w) \,\|\, p(w)\big], \qquad w = \mu + \sigma \odot \varepsilon, \tag{1}$$

with the KL weight warmed up linearly, $\sigma = \mathrm{softplus}(\rho)$ and ρ initialised at -5 ($\sigma_0 \approx 0.007$), Adam 10^{-3}, gradient clipping, 50 posterior samples at test time. By default BBB trains for 150 epochs at $n \geq 2{,}000$ (4,800 steps at $n = 2{,}000$) and the same number of steps at $n = 500$.

The softness-matched control. For a method M, I fit a temperature T to a DNN's logits so that its mean predictive entropy on the *validation* set equals M's. No test or OOD data enters the fit. The reported quantity is $\Delta = \mathrm{AUROC}(M) - \mathrm{AUROC}(\mathrm{DNN}_T)$, where AUROC measures the separation of test from OOD images by predictive entropy. $\Delta > 0$ means M ranks OOD inputs better than a network of the same softness. I report both entropy- and MI-based AUROC where relevant. Comparisons are paired by seed, with bootstrap confidence intervals and sign counts (wins out of seeds). Wilcoxon tests are reported only with 10 seeds, since with 5 seeds they cannot fall below $p = 0.0625$. Unless stated otherwise, BBB predictions use 50 posterior samples. The

training-dynamics runs (§3.3, §3.6) use 20, which slightly under-estimates OOD entropy, so their Δ values are conservative.

3 Results on MNIST

3.1 At standard budgets, only ensembles beat the control

Table 1 compares each method with its own softness-matched DNN over 10 seeds. At $n = 500$ and 2,000, BBB is significantly *worse* than the control; at $n = 10{,}000$ it is much better, in all 10 seeds. §3.2 shows that this jump is not about data. Deep ensembles beat the control at every n by $+0.01$ to $+0.02$; against a stronger, budget-matched MAP network their edge mostly disappears at $n = 2{,}000$. MC dropout roughly ties the control, and last-layer Laplace is clearly below it.

n	method	Δ Fashion [95% CI]	Δ KMNIST [95% CI]
500	BBB	-0.046 [-0.063, -0.032]	-0.035 [-0.041, -0.028]
500	ensemble	$+0.014$ [0.003, 0.027]	$+0.010$ [0.008, 0.012]
500	MC dropout	-0.003 [-0.016, 0.009]	-0.019 [-0.021, -0.018]
500	last-layer Laplace	-0.072 [-0.088, -0.055]	-0.112 [-0.122, -0.102]
2,000	BBB	-0.024 [-0.048, -0.001]	-0.010 [-0.014, -0.007]
2,000	ensemble	$+0.015$ [0.007, 0.022]	$+0.015$ [0.013, 0.018]
2,000	MC dropout	$+0.007$ [0.002, 0.012]	-0.010 [-0.012, -0.007]
2,000	last-layer Laplace	-0.040 [-0.047, -0.032]	-0.043 [-0.047, -0.038]
10,000	BBB	$+\mathbf{0.048}$ [0.038, 0.059]	$+\mathbf{0.069}$ [0.065, 0.074]
10,000	ensemble	$+0.016$ [0.008, 0.026]	$+0.023$ [0.021, 0.025]
10,000	MC dropout	-0.010 [-0.018, -0.001]	-0.007 [-0.010, -0.003]
10,000	last-layer Laplace	-0.014 [-0.023, -0.006]	-0.012 [-0.016, -0.008]

Table 1: Each method vs a DNN softened to its own validation entropy (entropy AUROC difference, 10 seeds). Source: `experiments/E2_mnist_grid`.

On accuracy, BBB beats the budget-matched MAP network by 1.1 points at $n = 500$ and 0.5 points at $n = 2{,}000$ (9/10 and 10/10 seeds), and loses by 0.5 points at $n = 10{,}000$. About half of its apparent lead over the plain DNN at $n = 2{,}000$ is ordinary regularisation.

3.2 The advantage follows optimizer steps, not data

With fixed epochs, a larger n also means more optimizer steps: 23,550 at $n = 10{,}000$ vs 4,800 at $n \leq 2{,}000$. Figure 1 decouples the two. With the same data, 23,550 steps give $\Delta = +0.06$ to $+0.08$ in 5/5 seeds at every n, including $n = 500$. 4,800 steps give no advantage, including at $n = 10{,}000$. Over this range the training-set size barely matters.

3.3 Along training: entropy and MI diverge

To see what happens in between, I trained BBB for 50,000 steps at $n = 2{,}000$ with a *constant* learning rate and evaluated every 1,000 steps (Figure 2). With the default initialisation, the entropy-based Δ on KMNIST is negative for the first $\sim$3,000 steps, peaks at $+0.067$ to $+0.074$ around 9,000–14,000 steps (mean first-layer σ 0.65–0.82), and is negative again at 25,000–30,000 steps. The MI-based Δ does not fall: it is $+0.073$ at 10,000 steps, $+0.071$ at 30,000 and $+0.061$ at 50,000. The entropy advantage disappears because the model also becomes uncertain on in-distribution

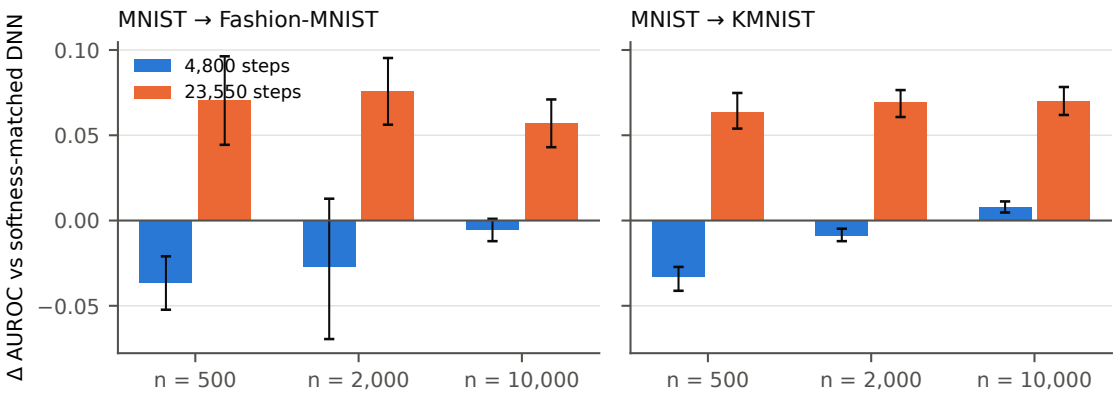


Figure 1: BBB vs its softness-matched DNN with the same training data and two step budgets (5 seeds, 95% CIs). Source: `experiments/E4_steps_vs_n`.

digits. Mean test entropy rises from 0.30 to 0.94 as accuracy falls from 0.93 to 0.89. The negative ELBO keeps improving throughout (143 at 4,000–5,000 steps, 6.8 at 24,000–25,000, 3.5 at 50,000). It drifts towards a posterior with $\sigma \approx 0.98$ in the hidden layers (0.93 in the last), accuracy 0.89 and ECE 0.19. That is consistent with Coker et al.'s analysis of wide mean-field BNNs [12], whose theorem covers a different setting (regression, odd activations). The entropy collapse is specific to $n = 2{,}000$: at 40,000 steps the entropy-based Δ is still $+0.033$ ($n = 500$) and $+0.060$ ($n = 10{,}000$) on KMNIST.

Two cautions about MI. First, a slightly larger initial σ (0.049) reaches the advantage sooner (3,000 steps instead of 6,000), but initialising $\sigma \geq 0.31$ prevents learning altogether: accuracy stays at chance in a local optimum with a far worse ELBO (9.0–9.2 vs 3.4–3.5). These chance-level models have the *largest* MI-based Δ (about $+0.10$). A positive MI Δ therefore does not by itself show a useful model. It also reflects disagreement between near-prior posterior samples on any unusual input. Second, these runs use a constant learning rate; the E4 runs (§3.2) used a cosine schedule, which freezes training wherever the budget ends.

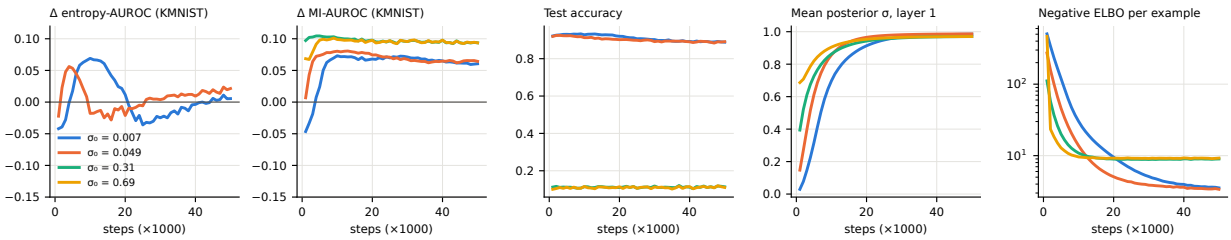


Figure 2: BBB training dynamics at $n = 2{,}000$ (constant learning rate; mean of 3 seeds per initial σ_0; 20 posterior samples). Runs with $\sigma_0 \geq 0.31$ never learn; their entropy-based Δ (about -0.86) is below the plotted range. Source: `experiments/E6_dynamics`.

3.4 An optimizer clock

Why do steps matter more than data? In the E9 runs (constant learning rate 10^{-3}, $\rho_0 = -5$), the mean first-layer ρ (with $\sigma = \text{softplus}(\rho)$) follows nearly the same path at every training-set size. It

is −3.54, −3.52 and −3.62 after 1,000 steps, −1.88, −1.85 and −1.92 after 3,000, and −1.01, −0.99 and −1.09 after 5,000, for $n = 500$, 2,000 and 10,000. With σ tiny at initialisation, the gradient on ρ is dominated by the KL term. Adam normalises its updates, so each step moves ρ by roughly the learning rate whatever the KL weight $1/N$ is. Unlu and Aitchison [13] point out this step bound for mean-field variances. If this explanation is right, BBB's state, and hence its OOD behaviour, is set by the learning rate times the number of steps and the initial σ, not by N. I have not yet tested this directly (by varying the learning rate, or by replacing Adam with SGD, where $1/N$ should matter), so it is a hypothesis.

3.5 What produces the advantage

Noise in both hidden layers. Taking BBB models trained for 23,550 steps, I switched posterior noise off layer by layer. Noise in the first and second layers together keeps 73–82% of Δ. Noise in any single layer keeps essentially none (first layer: 16% on Fashion-MNIST, −26% on KMNIST). The posterior-mean network on its own is a worse OOD detector than the DNN. The advantage comes from noise compounding across the hidden layers, which a last-layer method cannot produce.

Not a pixel shortcut. Many pixels are constant (always background) across a training subset: on average 192, 148 and 104 at $n = 500$, 2,000 and 10,000. A model-free rule, "flag any image with ink on such a pixel", separates MNIST from Fashion-MNIST and KMNIST with AUROC 0.967–1.000 over 15 training subsets (means 0.990–1.000), a real weakness of these benchmarks. But the models do not rely on it. Resetting those pixels to background in the OOD images leaves every method's AUROC and Δ essentially unchanged (BBB at $n = 10{,}000$: $+0.057 \to +0.055$ and $+0.070 \to +0.070$). The growth of σ is also global, not concentrated on those pixels (ratio 1.0–1.24).

3.6 Choosing when to stop

Neither in-distribution criterion picks a good checkpoint. Minimum held-out NLL stops at 1,000–3,000 steps, before the advantage appears; minimum −ELBO stops at the end of training. I tested a rule that needs no real OOD data. I built three synthetic outlier sets from the validation images (uniform noise, flipped digits, pixel-permuted digits) and picked the checkpoint that best separates validation images from them by MI (Table 2; 12 runs, $n \in \{500, 2{,}000, 10{,}000\}$). It reaches 73% of the best achievable entropy-based advantage (the "oracle"), but simply stopping at a fixed 8,000 steps reaches 72%. Scoring the last checkpoint by MI instead of entropy, with no selection at all, does as well (+0.049). The oracle is a maximum over 40 checkpoints and is optimistic. These runs also share configurations and seeds with §3.3. The proxy rule therefore does not beat trivial alternatives here.

rule	mean step	$\bar{\Delta}$ (entropy)	fraction of oracle
oracle (uses real OOD; optimistic)	10,917	+0.067	1.00
synthetic-outlier proxies (MI)	10,417	+0.049	0.73
fixed stop at 8,000 steps	8,000	+0.048	0.72
fixed stop at 10,000 steps	10,000	+0.047	0.71
minimum −ELBO	39,417	+0.034	0.52
minimum validation NLL	2,250	−0.032	−0.49

Table 2: Stopping rules ($\bar{\Delta}$ = mean of Fashion-MNIST and KMNIST entropy-based Δ at the chosen checkpoint; 12 runs). Source: `experiments/E9_selection`, `results/E9/`.

4 CIFAR-10

I repeated the dynamics experiment on CIFAR-10 (n = 10,000, no augmentation) with a small CNN (three 3×3 conv layers of 32, 64 and 128 channels, then 256 and 10 units), every layer Bayesian. OOD sets were SVHN (10,000 images) and CIFAR-100. No comparable advantage appeared (Figure 3). BBB's peak Δ on SVHN was −0.044, +0.006 and −0.056 in the three seeds, and on CIFAR-100 at most +0.031, early in training. Even a 5-member deep ensemble lost to a single softness-matched DNN on SVHN (AUROC 0.727–0.740 vs 0.759–0.807), while winning on CIFAR-100 by +0.02 to +0.03. Scored by MI instead, BBB was *worse* on SVHN: MI-AUROC ranged from 0.24 to 0.60 across checkpoints, often below chance. The drift towards the prior did replicate. The large fully connected layer's σ reached 0.996, the ELBO kept improving, and accuracy fell from 0.67–0.68 to 0.61–0.63. The first conv layer's σ stayed small (0.04).

To separate architecture from data, I trained the MNIST study's MLP on flattened CIFAR-10 images at n = 2,000 and 10,000 (3 seeds each). Its deterministic version reaches 38% and 46% test accuracy. BBB again showed no advantage on SVHN (peak Δ −0.031 to +0.030). On CIFAR-100 it gained +0.030 to +0.046 for the first few thousand steps, similar to the CNN's +0.024 to +0.031. Then it collapsed towards the prior (first-layer $\sigma \to$ 0.997–0.999, accuracy falling to 0.10–0.29). Much larger CIFAR-100 "gains" at n = 2,000 (+0.09 to +0.21) appear only after collapse, at near-chance accuracy, and are not meaningful. My pre-registered decision rule counted these runs as support for an architectural explanation, but it had no accuracy condition, and the qualifying gains are barely above its threshold and no larger than the CNN's. On natural images, neither architecture shows the MNIST-sized advantage (+0.06 to +0.08 on both OOD sets). The evidence points more to the data, or to the much harder task (37–47% accuracy), than to the CNN's structure.

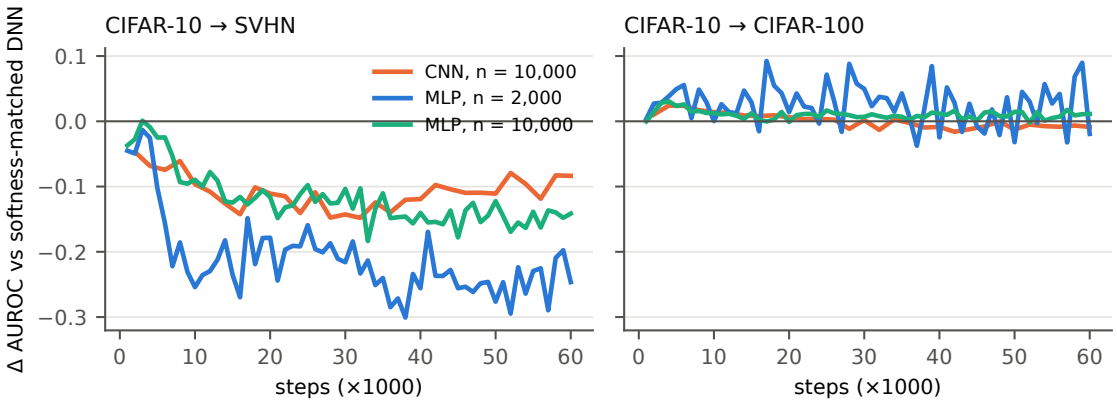


Figure 3: BBB vs its softness-matched network on CIFAR-10 over training (mean of 3 seeds per curve; CNN from all layers Bayesian, MLP = the MNIST architecture on flattened images). Sources: `experiments/E7_cifar`, `experiments/E7b_cifar_mlp`.

5 Related work

Coker et al. [12] prove that the optimal mean-field posterior predictive approaches the prior predictive as width grows; the drift in §3.3 is consistent with that effect along the optimisation path of a standard-width ReLU classifier, a setting their theorem does not cover. Unlu and Aitchison [13] note that mean-field variances are often unconverged when training stops; here that state is exactly where BBB looks like a tempered DNN. Izmailov et al. [14] show that weights on always-zero pixels keep their prior under HMC and hurt accuracy under covariate shift; I find no such pixel-specific effect in BBB's OOD detection. Ovadia et al. [4] include temperature-scaled baselines and find ensembles strongest under shift. Odgers et al. [15] show that mean-field VI can overestimate predictive variance on in-distribution points, which matches the rising in-distribution entropy in §3.3. Trippe and Turner [16] describe overpruning, where variational BNNs switch off units by sending their weights to the prior. Fortuin et al. [17] show that isotropic Gaussian priors are often poor choices for neural networks. Henning et al. [5] question whether exact BNN posteriors are intrinsically good OOD detectors. Raina and Schmah [18] report Bayesian advantages for OOD detection from small training sets; the results here suggest checking training length and softness before attributing such advantages to the posterior.

6 Limitations

- **Scope.** The positive results come from small MLPs on MNIST-family data. The CIFAR-10 models were small and underfit (37–68% accuracy, no augmentation), so the CIFAR result says little about well-trained networks.
- **Prior.** The $\mathcal{N}(0, 1)$ prior is badly misspecified for layers whose weights have scale ~ 0.05 [17], so a drift towards it is expected to hurt fit. I have not swept the prior scale.
- **Schedule and samples.** The dynamics runs use a constant learning rate and 20 posterior samples; the fixed-budget runs use a cosine schedule and 50 samples.
- **Statistics.** Dynamics results rest on 3 seeds per setting, and the selection analysis reuses those configurations and seeds.

- **No exact posterior.** Without HMC or similar, I cannot say whether any of this reflects Bayesian behaviour or only mean-field VI under Adam.

7 Conclusion

For Bayes by Backprop on MNIST-family data, "knowing what it doesn't know" is not a fixed property of the method. It depends on how many optimizer steps were taken and on which uncertainty score is reported. At typical budgets BBB behaves like a temperature-scaled network. With longer training its mutual-information signal separates OOD inputs well, but the same signal appears in models that never learned, and entropy-based scores can lose the advantage as the model underfits. Comparisons between BBB and deterministic or ensemble methods should report the training budget, the score and a softness-matched baseline.

Reproducibility. Code: `bnnstudy/`; every command: `COMMANDS.md`; every number in this report: `CLAIMS.md`; pre-registered protocols with results: `experiments/`.

A KL scaling

Dividing the KL term in Eq. 1 by the number of mini-batches instead of by N is an easy slip when the cross-entropy is averaged per batch. It over-weights the KL term by $N/\lceil N/B \rceil \approx B$ (62.5× at $N = 2{,}000$, $B = 64$). In a 3-seed pilot it cost BBB 3.2 points of accuracy (0.904 vs 0.936) and raised ECE from 0.022 to 0.185.

B Pre-registered predictions and outcomes

Each protocol (`experiments/<id>/protocol.md`) was committed before its runs. Of the 29 entries below, 9 were confirmed, 16 falsified, 3 had mixed outcomes and 1 had no outcome. (E5's P1 held by construction and is omitted.) The falsified ones are what moved the study from "is the advantage just softness?" to the training-path account.

exp.	prediction	outcome
E1	P0: new pipeline reproduces old numbers (± 0.02 acc, ± 0.03 AUROC)	acc/ECE yes; AUROC off by 0.035
E1	P1: correct KL weight raises BBB accuracy	confirmed
E1	P2: BBB within 0.02 AUROC of softness-matched DNN	falsified (Fashion)
E2	P1: BBB never beats the control	falsified ($n = 10{,}000$)
E2	P2: ensembles beat the control	confirmed
E2	P3: MC dropout and Laplace within 0.02 of the control	dropout yes; Laplace falsified
E2	P4: larger KL weight lowers accuracy, raises KMNIST MI-AUROC	confirmed
E3	P1: BBB and budget-matched MAP equally accurate (< 0.005)	falsified (narrowly)
E3	P2: BBB never beats softness-matched MAP	falsified ($n = 10{,}000$)
E4	P1: the gain follows optimizer steps, not n	confirmed
E4	P2: dead-pixel weights have $> 5\times$ the σ of live ones	falsified
E4	P3: $n = 500$ with long training shows the gain	confirmed
E5	P2: removing dead-pixel ink halves BBB's gain	falsified
E5	P3: removing it shrinks the ensemble's gain	falsified
E5	P4: the DNN's AUROC drops less than BBB's when the ink is removed	falsified (neither drops)
E6	P1: Δ rises with σ	falsified (peaks, then falls)
E6	P2: Δ plateaus	falsified (transient)
E6	P3: larger σ_0 reaches the gain sooner	falsified ($\sigma_0 \geq 0.31$ never learns)
E6	P4: ELBO still improving at 4,800 steps	confirmed
E6b	P1: first-layer noise alone keeps $\geq 70\%$	falsified
E6b	P2: last-layer noise alone keeps $< 30\%$	confirmed
E9	P1: combined proxies reach $\geq 70\%$ of oracle; NLL $\leq 20\%$	confirmed, but a fixed 8k-step stop does as well
E9	P2: one proxy alone reaches $\geq 70\%$	falsified (67%)
E9	P3: peak step rises with n	confirmed by mean; not robust
E7	P1: window on SVHN with a CNN	falsified
E7	P2: window closes	no outcome (no window)
E7	P3: proxy selection beats NLL	falsified
E7	P4: ensemble beats the control on SVHN	falsified
E7b	rule: architecture vs data	"architecture" by the rule; weak

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